



EE 583: Smart Grids

Course Code	EE 583
Course Name	Smart Grids
L-T-P-C	3-0-0-3
Prerequisites	EE-303 (Power systems)
Intended for	UG/PG
Distribution	Elective for B. Tech (EE), M.S., M. Tech. & Ph.D.
Approval	recommended by the 37th BoA (Oct 2020)

Course Contents

- **Introduction:** Smart Grid: Concept, architecture, standards and protocols, Smart Grid in Indian Grid context. [3 Lectures]
- **Power System Operations:** Load flow for smart grids, voltage stability assessment, and state estimation. [6 Lectures]
- **Communication Infrastructure:** [4 Lectures]
 - Communication standards, requirements for data links, data quality, cyber security: state of the art, risks and mitigation
- **Wide Area Measurement Systems (WAMS):** - Phasor measurement units (PMU), WAMS architecture, applications of WAMS for power system operation improvement: advantages and disadvantages. [3 Lectures]
- **Integration of Renewable Energy Sources and Energy storage:** Renewable energy sources, penetration and variability issues, environmental implications, demand response, electric vehicles, energy storage techniques: battery, pumped hydro, modelling of storage devices. [8 Lectures]
- **Smart Devices:** FACTS, STATCOM, HVDC, fault current limiters. [5 Lectures]
- **Protection and Security:** Intelligent protection, contingency analysis and classification, security, outage management, remedial action schemes, special protections schemes. [5 Lectures]
- **Meters and Sensors:** Hardware, demand side integration, communication standards and protocols, smart meters, automatic meter reading (AMR), advanced metering infrastructure (AMI). [4 Lectures]
- **Microgrids:** System operation, consumer energy management. [3 Lectures]
- **Hardware in Loop (HIL) testing:** HIL requirements, advantages and disadvantages. [1 Lecture]

Text Books

1. S. F. Bush, **Smart Grid: Communication-enabled intelligence for the electric power grid**, John Wiley and Sons, Ltd., 2014.
2. I. S. Jha, S. Sen, R. Kumar, D. P. Kothari, **Smart Grid Fundamentals & Applications**, New Age International Publishers, 2019.

References:

1. J. Momoh, **Smart Grid: Fundamentals of design and analysis**, John Wiley and Sons, Ltd., 2012.
2. B. M. Buchholz, Z. Styczynski, **Smart Grids - Fundamentals and Technologies in Electricity Networks**, Springer, 2014.
3. C. W. Jennings, **The Smart Grid: Enabling Energy Efficiency and Demand Response**, Fairmont Press Inc., 2009.
4. N. Hatziargyriou, **Microgrids: Architectures and Control**, John Wiley and Sons, Ltd., 2014.